

NAME: _____

PERIOD: _____

When Is the Mean Nearly Normal?

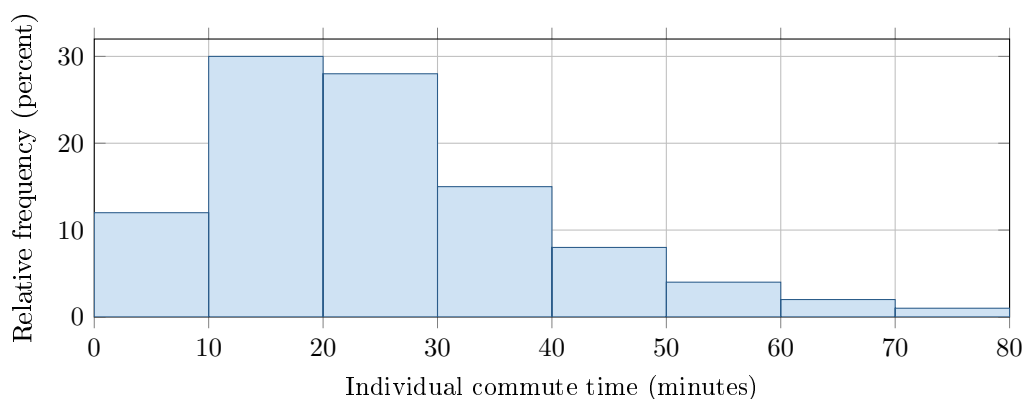
AP Statistics · Unit 2 · Topic 2.12

[STAR] THE PROBLEM

Individual commute times in a very large city are strongly right-skewed, with population mean $\mu = 25$ minutes and standard deviation $\sigma = 15$ minutes. Consider independent random samples.

Rae: “Any average is approximately normal, even for $n = 4$.”

Sol: “With this long right tail, $n = 4$ may be too small; $n = 40$ is more defensible.”



FIRST TAKE — one sentence before you work the parts

Which would you expect to smooth out the long commute times more: averaging 4 people or 40? Give one reason.

[BOOK] CLT: SHAPE, SIZE, AND THE RIGHT DISTRIBUTION (keep on the page)

The sampling distribution of $\bar{x}$ contains the sample means from all possible random samples of one size n . For independent observations, $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$. The **central limit theorem** says that when n is sufficiently large, the sampling distribution of $\bar{x}$ is approximately normal even if the population is not. Stronger population skew generally requires a larger n ; a small sample can retain that skew. The CLT concerns sample means, not individual observations.

Work (a)–(c).

DOK 1 (a) Identify the population shape and state the population mean and standard deviation with units.

DOK 2 (b) For $n = 4$ and $n = 40$, calculate the mean and standard deviation of the sampling distribution of $\bar{x}$.

DOK 3 (c) In 3–4 sentences, decide whose claim is better supported. Use the population shape, independence, and the two sample sizes. Explain which distribution the CLT describes and why using a normal curve for averages of 4 could be unreliable.

Frame: For 4 people, _____; for 40 people,
_____.

Frame: The CLT describes _____, so using a normal curve for 4 could
_____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.